A 10-Bit 500-MS/s Pipelined SAR ADC With Feedback Factor Compensation in 6-nm FinFET

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***Abstract*—This brief introduces an energy-efficient 10-bit 500-MS/s pipelined SAR ADC that uses feedback factor compensation in 6-nm FinFET technology. The design challenges of residue amplifier in FinFET technology are adequately addressed by incorporating feedback factor compensation. This includes a dynamic negative capacitance circuit at the virtual ground that compensates for the feedback factor and relaxes the requirements of the residue amplifier. This enables the use of an inverter-based residue amplifier that can achieve a high-speed operation of 500-MS/s at a low supply voltage of 0.9 V. The prototype ADC is fabricated in a 6-nm FinFET and occupies 0.014 mm^2. With a Nyquist input signal, it achieves an SNR of 54.2 dB and an SNDR of 53.6 dB, while consuming 2.7 mW from a 0.9 V supply voltage. This brief achieves a competitive Walden figure of merit (FoM) of 13.8 fJ/conv.-step.**

***Index Terms*—Analog-to-digital converter (ADC), 6-nm FinFET process, pipelined SAR ADC, feedback factor compensation, negative capacitance (NC), NC-assisted residue amplifier.**

I. Introduction

IN ADVANCED process nodes, e.g., sub-10 nm process, the level of integration can be significantly increased, but this introduces several challenges for ADC design. For instance, it becomes increasingly difficult to achieve high open-loop gain (A_{OL}) due to the reduction in supply voltage. In addition, design constraints arise from the significant increase in parasitic components. As shown in Table I, the advanced nodes of FinFET technology exhibit a significant increase in parasitic resistance R_P due to additional layers in the fabrication process [1]. An alternative is to utilize wider metal interconnects or additional fins; however, this comes at the cost of increased parasitic capacitance C_P. Consequently, designing using sub-10 nm process poses significant challenges.

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TABLE I
INTERCONNECT PARASITICS WITH TECHNOLOGY SCALING

	40nm	28nm	16nm	7nm	5nm
C_P	1	0.95	0.83	1.05	1.07
R_P	1	1.5	6	20	63

Due to the limited voltage headroom, a cascade approach has become inevitable. In response, a multi-stage amplifier, such as a ring amplifier, has been developed that achieves high energy efficiency by exploiting slew-based charging [2], [3], [4], [5], [6]. Nevertheless, the multi-stage amplifier requires precise stabilization techniques to ensure stability within the closed-loop configuration. An alternative approach is to use open-loop dynamic or G_m-based amplifiers for the residue gain. These have been shown to achieve high energy efficiency by circumventing the stringent requirements on the closed-loop residue amplifiers [7], [8], [9], [10], [11]. However, the open-loop configuration is susceptible to variations in process, voltage, and temperature (PVT), requiring additional calibration that increases the design complexity.

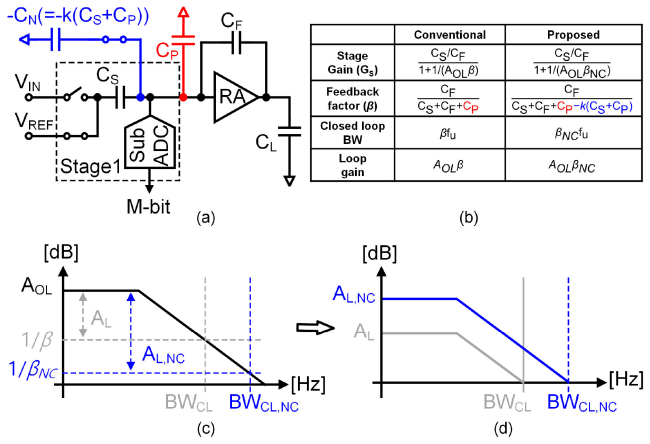
On the other hand, a feedback factor (β) compensation technique using a reference buffer or negative capacitor (NC) at the virtual ground of the residue amplifier has been proposed [12], [13]. This offers a clear advantage in terms of reducing the requirements on the main amplifier, resulting in an improved energy efficiency. However, the main amplifier of these ADCs still uses a high-performance amplifier with a telescopic OTA [13] or two-stage OTA [12]. In this brief, by using the dynamic NC for β compensation, a design optimization of the OTA is described to achieve an improved energy-efficiency. Implemented in a 6-nm FinFET process, a 10-bit 500-MS/s pipelined SAR ADC using a single-stage inverter-based OTA achieves 53.6 dB SNDR and 66.9 dB SFDR for Nyquist input, respectively, while consuming only 2.7 mW, resulting in a Walden figure-of-merit (FoM$_W$) of 13.8 fJ/conv.-step. The remaining of this brief is organized as follows. Section II describes the proposed pipelined SAR ADC, while Section III summarizes the measurement results. Finally, this brief ends with conclusions in Section IV.

II. Proposed Pipelined SAR ADC

A. β Compensation Technique

Fig. 1(a) presents the closed-loop residue amplification incorporating the β compensation technique [12]. During the amplification phase, the NC ($-C_N$) is connected to the virtual

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	Conventional	Proposed
Stage Gain (G_s)	$\frac{C_S/C_F}{1+1/(A_{OL}\beta)}$	$\frac{C_S/C_F}{1+1/(A_{OL}\beta_{NC})}$
Feedback factor (β)	$\frac{C_F}{C_S+C_F+C_P}$	$\frac{C_F}{C_S+C_F+C_P-k(C_S+C_P)}$
Closed loop BW	βf_u	$\beta_{NC} f_u$
Loop gain	$A_{OL}\beta$	$A_{OL}\beta_{NC}$

Fig. 1. (a) The closed-loop residue amplification with the proposed β compensation, (b) comparison of conventional and proposed scheme, Bode plot for (c) A_{OL} and (d) A_L.

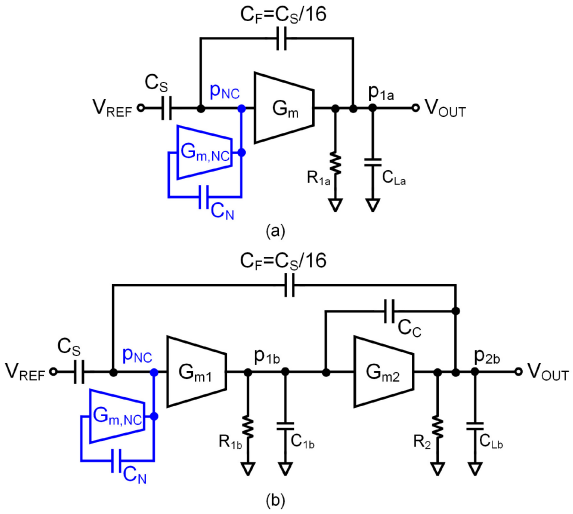


Fig. 2. Closed-loop NC-assisted residue amplifier ac model (a) 1-stage OTA and (b) 2-stage OTA.

ground to provide β compensation. This serves to mitigate the impact of C_P and C_S. As the compensation coefficient k approaches 1, ideal compensation is achieved ($\beta_{NC} = 1$). Fig. 1(b) illustrates that the β compensation enhances the loop gain (A_L) and the closed-loop bandwidth (BW_{CL}) residue amplifier by an x-factor of β_{NC}/β. Fig. 1(c) and (d) visualize the summary of (b) in the form of Bode plots for the A_{OL} and the adjusted A_L ($= A_{OL}\beta$).

B. Comparison of the NC-Assisted Residue Amplifiers

In prior work using a 28-nm CMOS process, the A_{OL} requirement could be reduced to 50 dB by applying β compensation technique and an inverter-based 2-stage OTA is used as a residue amplifier [12]. However, the use of 2-stage OTA requires additional compensation to ensure stability, which adds complexity to the design and limits its bandwidth, also resulting in higher power consumption. Fig. 2 depicts both the 1-stage and 2-stage OTAs with NC. The load capacitance (C_{La} and C_{Lb}) includes the parasitic capacitance resulting from metal routing associated with the residual amplifier,

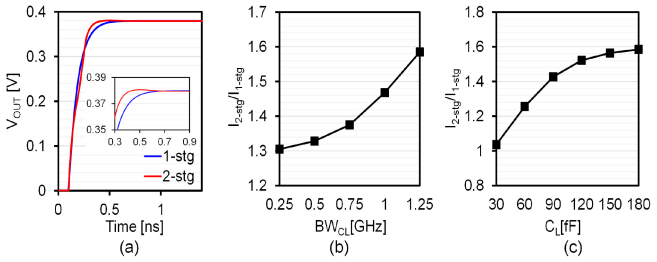


Fig. 3. (a) Step response of two amplifiers, simulated current consumption ratio of two amplifiers vs (b) BW_{CL} and (c) C_L.

	A_L	G_m @ 1-stg OTA (=G_m)	G_m @ 2-stg OTA (=G_{m1}+G_{m2})
w/o NC	25.4dB	26.2mS	34.1mS
w/ NC	37.5dB	8.0mS	15.6mS
X-factor		**3.3X**	**2.2X**

(a)

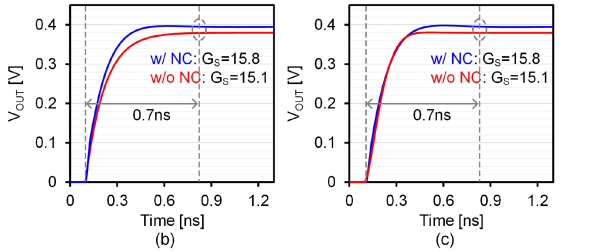


Fig. 4. Comparison of two schemes with NC (a) Summary of the simulated results, step response of (b) 1-stage OTA and (c) 2-stage OTA.

assuming C_{1b} to be one-third of C_{Lb}. Fig. 3(a) illustrates the step response of the two amplifiers, each with a similar A_{OL} (= 50 dB) and BW_{CL} (= 1.25 GHz). The absence of negative capacitance (NC) has resulted in an A_L of only 25 dB due to β, with the stage gain (G_S) failing to reach the desired 16×, thus resulting in a gain error.

A detailed examination of the zoomed-in waveform reveals that the 1-stage OTA exhibits a relatively improved stability. Fig. 3(b) shows that the current consumption of the 2-stage OTA increases compared to the 1-stage OTA as the required BW_{CL} rises. Fig. 3(c) indicates that the energy efficiency of the 2-stage OTA decreases with increasing C_L. Since the C_P in the FinFET process significantly enlarges the C_L, minimizing the number of amplifier stages is beneficial. Therefore, employing the 1-stage OTA as the main amplifier with NC to compensate for the deficient A_{OL} and BW_{CL} is a promising strategy to improve energy efficiency.

Fig. 4(a) depicts the A_L and required transconductance (G_m) for the combination of the NC and the two different OTAs. Although the two OTAs depicted in Fig. 3 necessitate a G_m of 26.2 mS and 34.1 mS, respectively, the A_L without the NC is limited to a 25.4 dB, and the G_S turns out to be insufficient (∼15.1). The NC-assisted OTA boosts the A_L to 37.5 dB, thereby achieving the resolution requirements for the subsequent stage (6 bits). The NC-assisted OTA achieves the required G_S (=15.8), while reducing the G_m of the main OTA to 8 and 15.6 mS, respectively. The requirements of the 1-stage and 2-stage OTAs can be reduced by 3.3× and 2.2×, respectively. As can be seen, the 1-stage OTA is 1.5 times more efficient than the 2-stage OTA. In the case of the 2-stage

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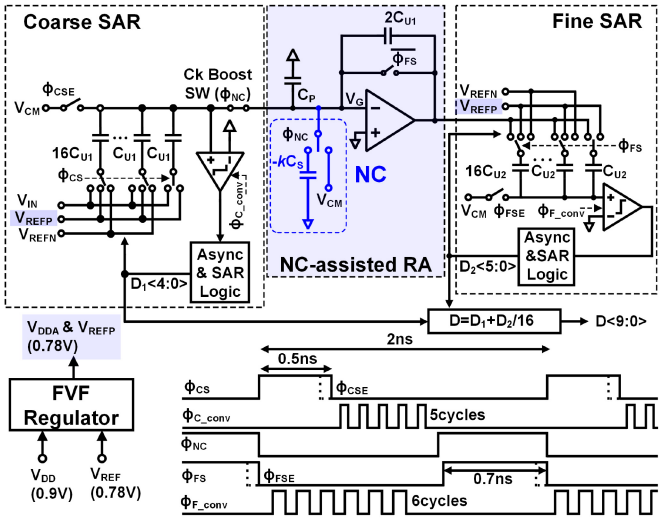


Fig. 5. Schematic and timing of proposed pipelined SAR ADC.

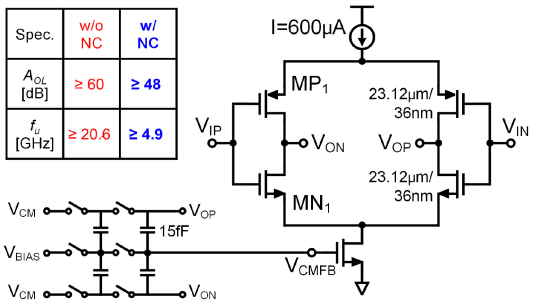


Fig. 6. Schematic of the main amplifier and its specification.

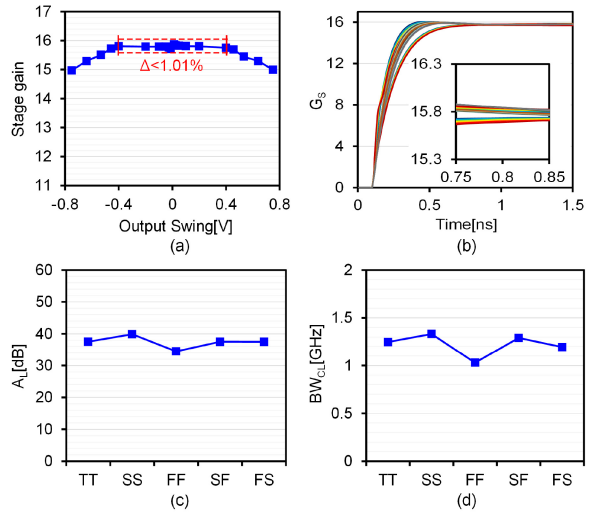


Fig. 7. Proposed dynamic NC assisted residue amplifier's simulation results, (a) Stage gain vs Output swing, (b) Step-response vs corner (TT, SS, FF, FS, SF) and temperature (0°C - 80°C), (c) A_L, and (d) BW_{CL}.

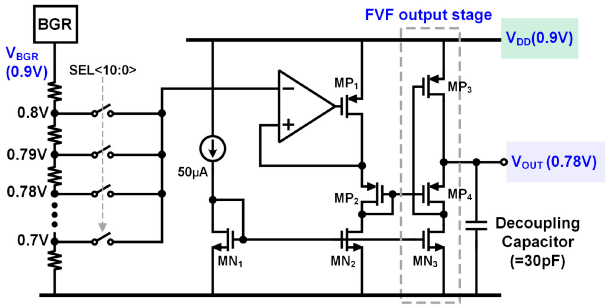


Fig. 8. Schematic of the on-chip regulator.

OTA, the pole introduced by NC (p_{NC}) is positioned between nodes p_{1b} and p_{2b}, additional current is required to drive p_{2b} at a high frequency to ensure stability.

C. ADC Implementation

Fig. 5 shows the schematic and timing diagram of the proposed pipelined SAR ADC. The C_{U1} of the 5-bit coarse SAR is 8.375 fF, and with a total sampling capacitance of 268 fF and a C_F of 16.75 fF, the G_S is 16×. The NC is of the dynamic type, as implemented in prior work [12]. The use of the 6-nm process introduces a significantly higher C_P at the virtual ground node compared to previous technology nodes, raising concerns about C_P variation and stability during β compensation. Consequently, the NC's compensation coefficient k is set to 0.8, with the NC designed to be 295 fF to compensate for 368 fF, which includes the C_P (≈100 fF) of the virtual ground node. This adjustment increases β from 0.059 to 0.24, relaxing the requirements for A_{OL} and f_u by a factor of 4.07. The inverter-based residue amplifier, shown in Fig. 6, must achieve the G_S of 16× within 700 ps, which requires the A_L of 36 dB and the BW_{CL} exceeding 1.2 GHz. The use of NC allows the required A_{OL} and f_u to be relaxed to 48 dB and 4.9 GHz, respectively. Fig. 7 presents the simulation results of the proposed dynamic NC-assisted residue amplifier. As shown in Fig. 7(a), it maintains approximately 16× gain with a variation of 1.01% within ±0.4 V differential output swing range. Fig. 7(b) illustrates the simulation results under temperature variations from 0 to 80°C across three process corners, which remains within a specified timing range within a maximum error of 0.5 LSB. The NC value shows minimal variation across the corners, with the highest value occurring at the SS corner and the lowest at the FF corner. Fig. 7(c) and 7(d) demonstrate that both the A_L and BW_{CL} are smallest at the FF corner due to the reduced NC value. Nevertheless, this reduction is not significant enough to pose any operational issues.

The 6-nm process exhibits significant C_P and R_P inherent to the implementation. In particular, the R_P in the lower metal layer is large enough to affect the performance of digital circuits composed of basic logic gates. To address this, the power supply is split, delivering 0.78 V to the analog section and 0.9 V to the digital section. Since the analog portion powers the residue amplifier and the V_{REFP}, a clean and stable voltage is required. To accomplish this, an on-chip flipped voltage follower (FVF) regulator is integrated. The 0.9 V supply powers both the FVF regulator and the digital components of the ADC, while the stabilized 0.78 V output is provided to the analog section. Fig. 8 shows the schematic of the on-chip regulator, which comprises a bias circuit, an error amplifier, and an FVF output stage [15]. The FVF output stage enables fast transient performance by leveraging a local feedback loop to reduce output impedance and enhance current

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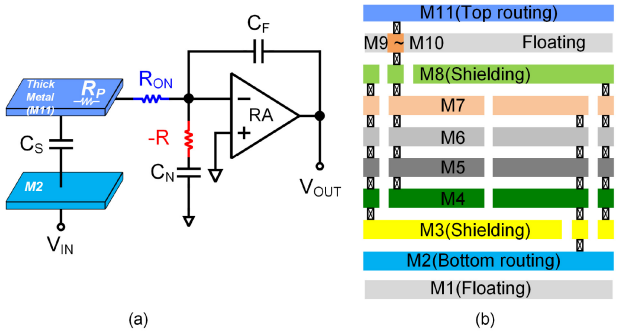


Fig. 9. (a) Routing of the C-DAC and Residue amplifier at virtual ground node and (b) encapsulated MOM capacitor layout and routing.

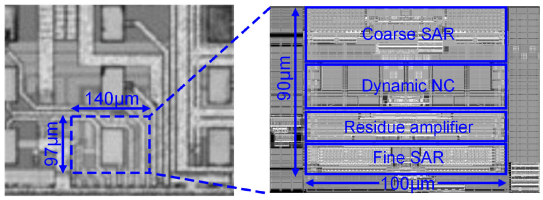


Fig. 10. Chip micrograph.

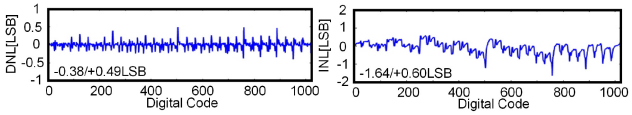


Fig. 11. Measured DNL/INL.

sourcing capability. A 30 pF decoupling capacitor is placed in the available idle space to ensure stable operation.

As shown in Fig. 9(a), the R_P of the virtual ground node must be minimized, as it can induce fluctuations in the R_{ON} used to compensate for the negative resistance (-R) of the NC. Fig. 9(b) illustrates the encapsulated MOM capacitor and its routing for the unit capacitor, which simplifies the application of the concept in Fig. 9(a). The unit capacitance is formed using metal layers M4 through M7, with bottom routing to M2 and top routing to M11. To reduce C_P between the bottom plate (M4) and routing (M2), M3 is used as a shield, while the lower metal layers M1 and M0 are left unused and floating, as lower layers typically exhibit higher R_P. To minimize the R_P of the top plate, thick metal (M11) is employed, effectively reducing R_P but at the cost of increased C_P. To address this, shielding is provided using M8, and layers M9 through M10 are left floating. Additionally, the capacitors are encapsulated by M4 through M7 to reduce capacitive coupling between adjacent capacitors. All shielding layers are grounded to maintain a stable potential.

III. Measurement Results

The proposed 10-bit 500-MS/s ADC prototype is fabricated in a 6-nm FinFET process. It occupies an active area of 0.014 mm^2, as shown in Fig. 10, where an active area excluding the integrated regulator and decoupling capacitor is only 0.009 mm^2. As shown in Fig. 11, the measured differential nonlinearity (DNL) and integral nonlinearity (INL) are −0.38/+0.49 and −1.64/+0.60, respectively. Fig. 12 shows

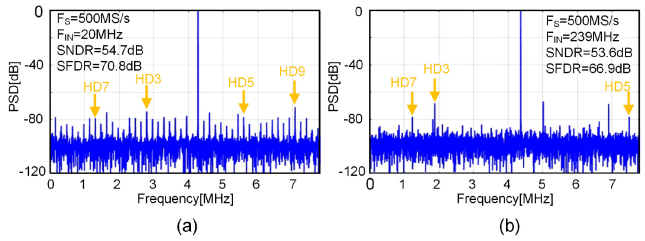


Fig. 12. Measured output spectrum with (a) 20-MHz, (b) 239-MHz.

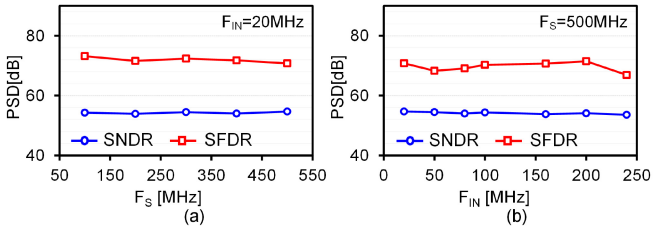


Fig. 13. Measured SNDR/SFDR vs (a) F_S, (b) F_{IN}.

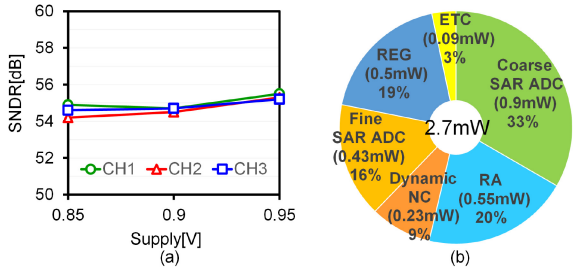


Fig. 14. (a) Measured SNDR with supply and chip variations and (b) power breakdown.

the measured power spectral densities (PSDs) decimated by a factor of 32. For the 20-MHz input, the SNDR and SFDR are 54.7 dB and 70.8 dB, respectively. For the near Nyquist input of 239-MHz, the measured SNDR and SFDR are 53.6 dB and 66.9 dB, respectively. Fig.13(a) shows the measured performance for different sampling frequencies at the 20-MHz input, where the variations of SNDR and SFDR are in the range of 0.8 dB and 2.4 dB, respectively. Fig. 13(b) demonstrates the measured performance for different F_{IN} at a sampling rate of 500-MS/s, which confirms that the performance remains stable up to the Nyquist frequency. By using the integrated regulator, the measured results for the supply variations of the three chips remain stable, as shown in Fig. 14(a). Fig. 14(b) shows the power breakdown of the ADC. The overall power consumption is only 2.7 mW. By using only 9% of the total power consumption with the dynamic NC, the power consumption of the residue amplifier can be reduced to 0.55 mW (∼20%).

Table II provides a performance summary and comparison with prior state-of-the-art works fabricated in a 28-nm process [12], or below [3], [5], [6], [14]. The use of an external reference regulator further enhances the FoM_W (= 9.6 fJ/conv.-step), as eliminating the power consumption of the integrated regulator and the use of a low-noise external regulator (LT3045) allow an improvement in SNDR. Compared to [12], this brief achieves both a better FoM_W

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TABLE II
PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART WORKS

	This work		A-SSCC' 23 Kwon [12]	ISSCC' 21 Baek [14]	ISSCC' 24 Lim [6]	ISSCC' 19 Hershberg [3]	JSSC' 22 Lagos [5]	
Process[nm]	**6**		28	7	8	16	16	
Architecture	**Pipelined SAR**		Pipelined SAR	SAR + TI-DSM	Pipelined SAR	Pipeline	Pipelined SAR	
Residue Amplifier	**Inverter-based 1-stage OTA**		Inverter-based 2-stage OTA	No	Ring Amplifier	Ring Amplifier	Ring Amplifier	
Amplifier Auxiliary technique	**NC-assisted β Compensation**		NC-assisted β Compensation	Residue path Replica	Dynamic bias	Gain calibration	Dither-based Gain calibration	
Resolution[bits]	**10**		11	12	12	11	12	
F_S[MS/s]	**500**		360	600	400	600	500	
Regulator	**Yes**	**No**	No	No	No	No	Yes	No
Area[mm^2]	**0.014**	**0.009**	0.016	0.037	0.017	0.037	0.047	0.015
Supply[V]	**0.9**	**0.78(A)/0.9(D)**	0.95	0.8	0.9	0.85	0.9	
Power[mW]	**2.7**	**2.1**	3.9	13	2.1	6	3.3	2.8
SNDR[dB]	**53.6**	**54.5**	58	55.3	62.8	60.2	62.3	62.9
SFDR[dB]	**66.9**	**68.3**	77.9	67.8	68.6	78.3	75.5	75.5
*FoM_W[fJ/c.-s]	**13.8**	**9.6**	16.7	45.6	4.6	12.0	6.2	4.9

*FoM_W = Power/($2^{ENOB} \times 2 \times BW$)

with a higher operation speed (1.38×). Compared to an ADC that does not use a residue amplifier in a 7-nm FinFET [14], this brief achieves significantly improved energy efficiency at similar resolution and speed. Compared to ADCs with FinFET process that use a ring amplifier [3], [5], [6], this brief achieves competitive energy efficiency and operation speed.

IV. CONCLUSION

This brief presents a pipelined SAR ADC featuring a dynamic NC-assisted residue amplifier fabricated in a 6-nm FinFET process. The dynamic NC enables high energy efficiency by exploiting a single-stage inverter-based OTA, required for low supply voltage operation. The ADC prototype achieves a competitive FoM_W of 13.8 fJ/conv.-step when operating at 500-MS/s, and is robust against PVT variations due to its closed-loop configuration. The combination of a single-stage OTA with a dynamic NC represents a promising solution for energy-efficient pipelined SAR ADCs in advanced process nodes.

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